Fast Rectifiers

UF4001 - UF4007

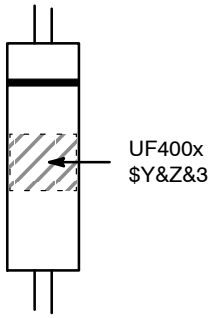


AXIAL LEAD / DO−41
CASE 017AH

Features

- Low Forward Voltage Drop
- High Surge Current Capability
- High Reliability
- High Current Capability
- Glass−Passivated Junction
- These are Pb−Free Devices

MARKING DIAGRAM



UF400x = Specific Device Code
x = 1/2/3/4/5/6/7
$Y = **onsemi** Logo
&Z = Assembly Plant Code
&3 = 3−Digit Data Code (Year & Week)

(Color Band Denotes Cathode)

ABSOLUTE MAXIMUM RATINGS

(Values are at T_A = 25°C unless otherwise noted.)

Symbol	Parameter	Value							Unit
		UF 4001	UF 4002	UF 4003	UF 4004	UF 4005	UF 4006	UF 4007	
V_{RRM}	Maximum Repetitive Reverse Voltage	50	100	200	400	600	800	1000	V
$I_{F(AV)}$	Average Rectified Forward Current .375" Lead Length at T_A = 75°C	1.0							A
I_{FSM}	Non−Repetitive Peak Forward Surge Current 8.3 ms Single Half−Sine−Wave	30							A
T_{STG}	Storage Temperature Range	−65 to +150							°C
T_J	Operating Junction Temperature	−65 to +150							°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

THERMAL CHARACTERISTICS

(Values are at T_A = 25°C unless otherwise noted.)

Symbol	Parameter	Value	Unit
P_D	Power Dissipation	2.08	W
$R_{\theta JA}$	Thermal Resistance, Junction−to−Ambient	60	°C/W
$R_{\theta JL}$	Thermal Resistance, Junction−to−Lead	30	°C/W

ORDERING INFORMATION

Device	Package	Shipping†
UF4001	Axial Lead (DO−41) (Pb−Free)	5000 / Tape & Reel
UF4002		
UF4003		
UF4004		
UF4005		
UF4006		
UF4007		

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

ELECTRICAL CHARACTERISTICS (Values are at T_A = 25°C unless otherwise noted.)

Symbol	Parameter	Test Conditions	Value							Unit
			UF4001	UF4002	UF4003	UF4004	UF4005	UF4006	UF4007	
V_F	Maximum Forward Voltage	I_F = 1.0 A	1.0				1.7			V
t_{rr}	Maximum Reverse Recovery Time	I_F = 0.5 A, I_R = 1.0 A, I_{RR} = 0.25 A	50				75			ns
I_R	Maximum Reverse Current at Rated V_R	T_A = 25°C T_A = 100°C	10 50							μA
C_T	Maximum Total Capacitance	V_R = 4.0 V, f = 1.0 MHz	17							pF

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

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TYPICAL PERFORMANCE CHARACTERISTICS

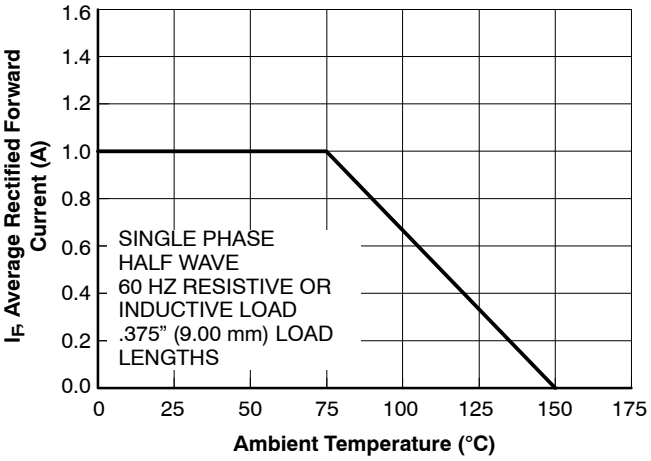


Figure 1. Forward Current Derating Curve

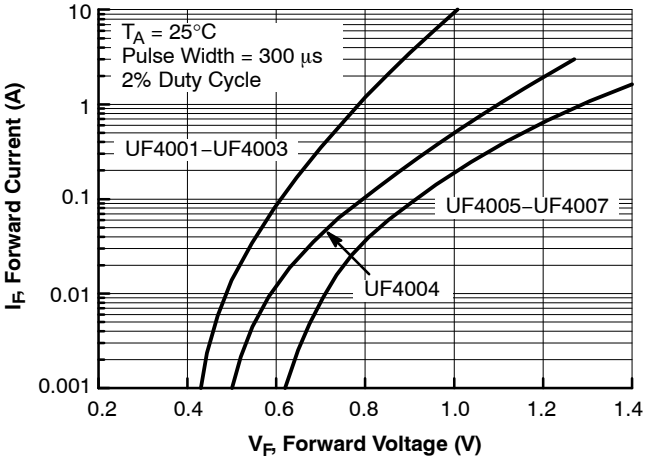


Figure 2. Forward Characteristics

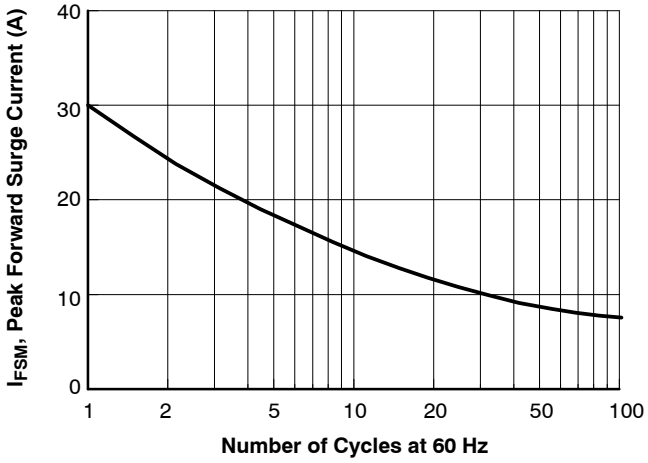


Figure 3. Non−Repetitive Surge Current

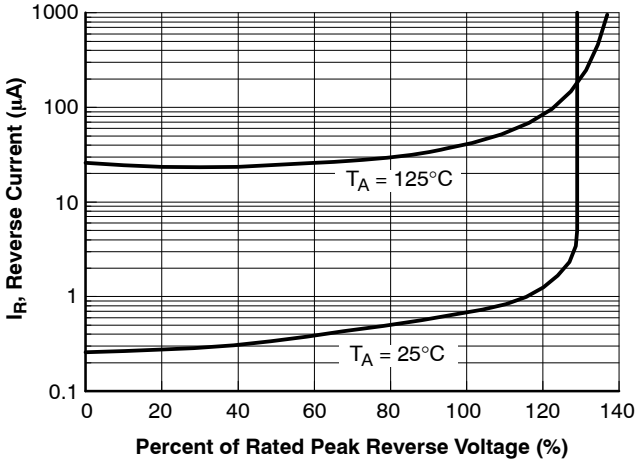


Figure 4. Reverse Characteristics

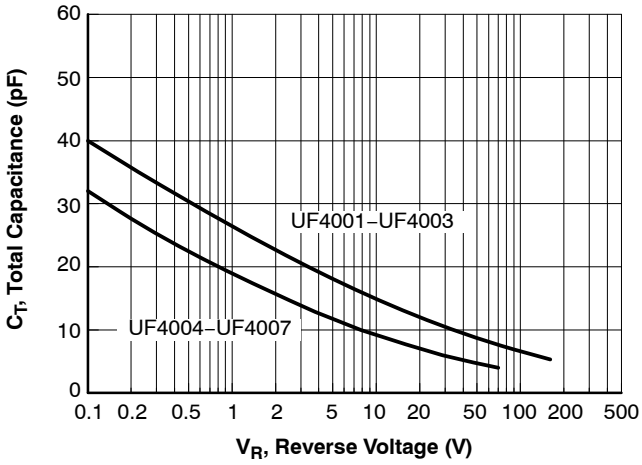
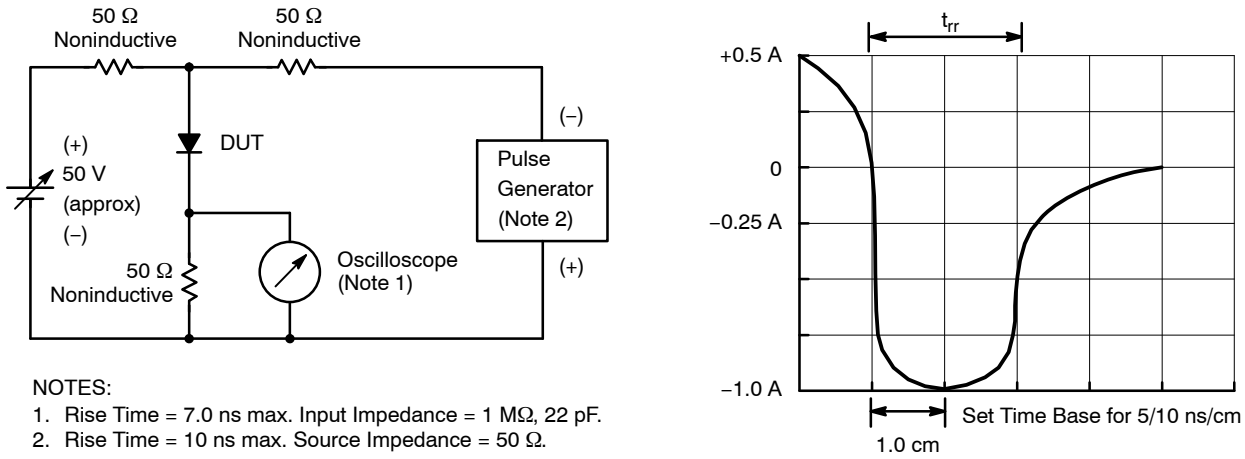


Figure 5. Typical Junction Capacitance

TEST CIRCUIT DIAGRAM

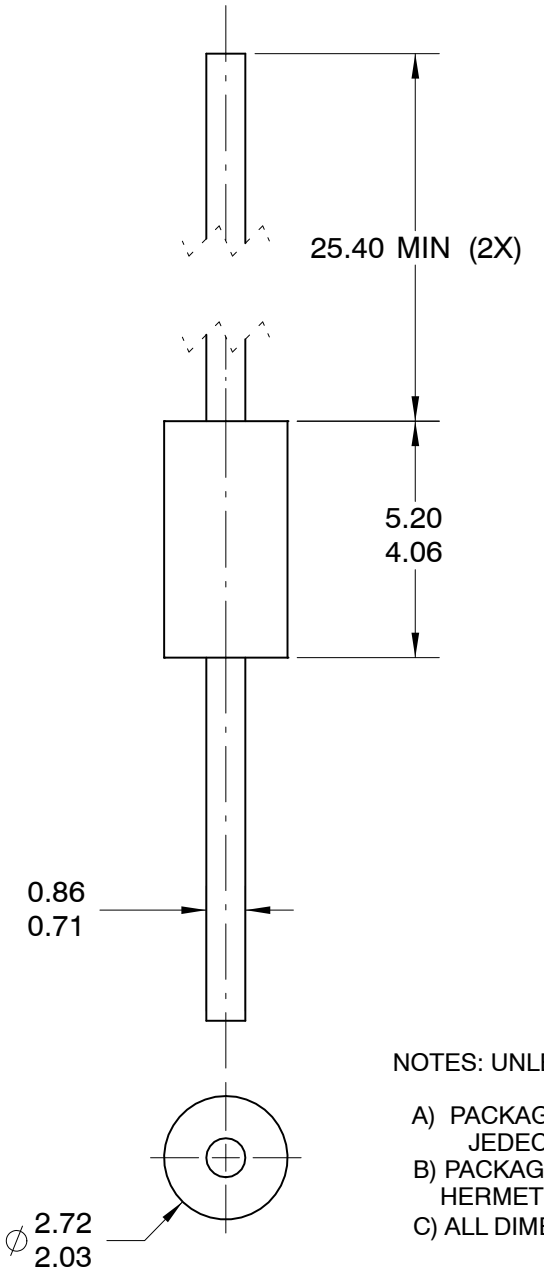


NOTES:
1. Rise Time = 7.0 ns max. Input Impedance = 1 MΩ, 22 pF.
2. Rise Time = 10 ns max. Source Impedance = 50 Ω.

Figure 6. Reverse Recovery Time Characteristic and Test Circuit Diagram

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ISSUE O

DATE 31 AUG 2016



NOTES: UNLESS OTHERWISE SPECIFIED

A) PACKAGE STANDARD REFERENCE:
JEDEC DO−204 VARIATION AL.
B) PACKAGE BODY CAN BE PLASTIC OR
HERMETICALLY SEALED GLASS.
C) ALL DIMENSIONS ARE IN MILLIMETERS.

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